Math 54 — Week 6

Solutions

P1.

Idea. One row reduction gives all three bases.

1. $R_2 + R_1$: $(0, -2, -5, -3)$. $R_3 - \frac{3}{2}R_1$: $(0, 2, 5, 3)$. $R_3 + R_2$: zero row. The echelon form, and the reduced echelon form after scaling and clearing above the pivots, are
$$\begin{bmatrix} -2 & 4 & -2 & -4 \\ 0 & -2 & -5 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 5/2 & 3/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$
2. $\operatorname{Col} A$: pivot columns 1, 2 of A: $(-2, 2, -3)$, $(4, -6, 8)$.
3. $\operatorname{Row} A$: $(-2, 4, -2, -4)$, $(0, -2, -5, -3)$.
4. $\operatorname{Nul} A$: $x_1 = -6x_3 - 5x_4$, $x_2 = -\frac{5}{2}x_3 - \frac{3}{2}x_4$: basis $(-6, -\frac{5}{2}, 1, 0)$, $(-5, -\frac{3}{2}, 0, 1)$ (or the multiples $(-12, -5, 2, 0)$, $(-10, -3, 0, 2)$).
5. Rank 2, nullity 2, $2 + 2 = 4 = n$.

P2. Three vectors in $\mathbb{R}^3$: by the basis theorem it is enough to check independence.
$$\det \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \\ -1 & 0 & 4 \end{bmatrix} = 1(4 - 0) - 2(0 + 3) + 1(0 + 1) = -1 \neq 0.$$
Yes, a basis.

P3.
$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \\ 2 & 5 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$
Pivot columns 1, 2: basis $\{(1, 0, 2), (2, 1, 5)\}$, $\dim H = 2$. (Indeed $(0, -1, -1) = -2(1, 0, 2) + (2, 1, 5)$.)

P4.

Idea. Turn $p(1) = 0$ into an equation on the coefficients.

$p = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ and $p(1) = a_0 + a_1 + a_2 + a_3 = 0$, so $a_0 = -a_1 - a_2 - a_3$ and $p = a_1(t-1) + a_2(t^2-1) + a_3(t^3-1)$. So $\{t - 1,\ t^2 - 1,\ t^3 - 1\}$ spans the subspace. It is independent: the three polynomials have different degrees, so in a relation the highest-degree term with nonzero weight could not cancel. Dimension 3.

P5. (a) $\dim \operatorname{Nul} A = 8 - 5 = 3$. $\operatorname{Col} A$ is a 5-dimensional subspace of $\mathbb{R}^5$, so it is $\mathbb{R}^5$ (basis theorem). (b) No: $\operatorname{rank} A \leq 4$, so $\dim \operatorname{Nul} A = 6 - \operatorname{rank} A \geq 2$.

P6. Row reduce

$$[\,\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{e}_1\ \mathbf{e}_2\ \mathbf{e}_3\,] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 & 1 \end{bmatrix}:$$

$R_2 - 2R_1$ gives $(0, 1, -2, 1, 0)$; $R_3 - 3R_2$ gives $(0, 0, 6, -3, 1)$. Pivot columns 1, 2, 3: basis $\{(1, 2, 0), (0, 1, 3), (1, 0, 0)\}$.

P7.

1. (a) True: more vectors than a basis.
2. (b) False: they could be dependent (for example equal), spanning a line or $\{\mathbf{0}\}$.
3. (c) True: both equal the number of pivots.
4. (d) True: $\operatorname{Nul} A \neq \{\mathbf{0}\}$, so A is not invertible, so $\det A = 0$.